

Full-Stack AI-Powered Sentiment Analysis Dashboard

Task Overview:

Build a simple web application that allows users to input text, analyze its sentiment using an AI model, and display the results on a dashboard. The application should have:

- A **frontend** built with React or Svelte
 - A **backend** built with Node.js (Express) or Python (Flask/FastAPI)
 - AI-powered sentiment analysis using an NLP model (such as OpenAI API, Hugging Face's transformers, or TextBlob for Python)
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Requirements:

Frontend (React/Svelte)

1. A simple UI with:
 - A text input box where users can enter text.
 - A submit button to analyze sentiment.
 - A results section displaying the sentiment (e.g., Positive, Negative, Neutral) and confidence score.
 - A simple dashboard showing a history of past analyzed texts.
2. Sends requests to the backend API.

Backend (Node.js with Express or Python with Flask/FastAPI)

1. Exposes an API endpoint (`/analyze`) that:
 - Accepts a text string from the frontend.
 - Passes the text to an AI-powered sentiment analysis tool.
 - Returns the sentiment label and confidence score.

2. Stores analyzed texts in an in-memory store or database (SQLite/PostgreSQL/MongoDB/Redis).
3. Exposes another API endpoint (`/history`) to fetch past analyzed texts.

AI Integration

- Use a simple sentiment analysis model
 - Process the text input and classify sentiment as **Positive, Negative, or Neutral**.
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Bonus Features (Optional for Extra Credit)

- **User Authentication:** Users can log in and save their analysis history.
 - **Data Visualization:** Show sentiment trends over time using a chart library (e.g., Chart.js, D3.js).
 - **Deploy the App:** Host on Vercel/Netlify (Frontend) and Render/Heroku (Backend) or any cloud service provider/host machine you have access to.
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Expected Outcome

A working full-stack application where users can input text, receive AI-powered sentiment analysis results, and view a history of past analyses on a simple dashboard.